

FULL STACK WEB DEVELOPMENT COURSE CONTENT

Duration- 8-12 Weeks

WEEK - 01

Web Basics & HTML/CSS

- Introduction to the Internet, HTTP, Browsers
- HTML5: Elements, Forms, Tables, Media
- CSS3: Selectors, Flexbox, Grid, Responsive Design
- Intro to Developer Tools and VS Code
- Project: Build a personal portfolio website

WEEK - 02

JavaScript Essentials

- Variables, Data Types, Operators
- Functions, Arrays, Loops, Conditionals
- DOM Manipulation, Events, Forms
- Local Storage, JSON
- Project: Dynamic to-do list or calculator

WEEK - 03

Advanced JavaScript & Es6

- Es6+ Features: let/const, Arrow Functions, Destructuring
- Callbacks, Promises, Async/Await
- Modules and Classes
- Error handling and debugging
- Project: Weather App using Open Weather API

WEEK - 04

Version Control & Deployment

- Git and GitHub: Commits, Branching, Pull Requests
- GitHub Pages for frontend deployment
- Basic CLI commands
- Netlify & Vercel deployments
- Practice: Host your mini project online

WEEK - 05

React.js - Frontend Framework

- React Basics: Components, Props, State
- Functional vs Class Components
- useState, useEffect hooks
- Routing with React Router
- Project: Single Page App (e.g., movie search, recipe app)

WEEK - 06

Backend with Node.js & Express

- Node.js Intro & environment setup
- Express.js basics: Routing, Middleware
- RESTful APIs and CRUD operations
- Connecting to frontend
- Project: REST API for a product inventory

WEEK - 07

MongoDB & Mongoose

- NoSQL vs SQL, MongoDB overview
- CRUD Operations with MongoDB Compass
- Mongoose schemas and models
- API + DB Integration
- Practice: User data collection app

WEEK - 08

Authentication & Authorization

- User registration & login
- Password hashing with bcrypt
- JWT (JSON Web Token) Authentication
- Protected routes and sessions
- Project: Secure blog or dashboard system

WEEK - 09

Advanced Topics

- File Uploads using Multer
- Email sending (e.g., Nodemailer)
- Error handling and validation
- Deployment with Render / Railway / Heroku
- Environment variables and config

WEEK - 10-12

Capstone Project

- Real-world Full Stack Project (Ideas: eCommerce App, Job Board, Chat App, etc.)
- Full CI/CD pipeline and deployment
- Responsive UI + Admin Panel (optional)
- Presentation and Peer Review

Tools & Technologies

- Frontend: HTML, CSS, JavaScript, React, Bootstrap/Tailwind
- Backend: Node.js, Express.js
- Database: MongoDB with Mongoose
- Dev Tools: Git, VS Code, Postman, GitHub
- Deployment: Netlify, Vercel, Railway/Render

Course Deliverables

- Source Code Repositories
- Project Templates
- Weekly Challenges & Assignments
- Capstone Project with Report
- Course Completion Certificate



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